CS H Review Guide

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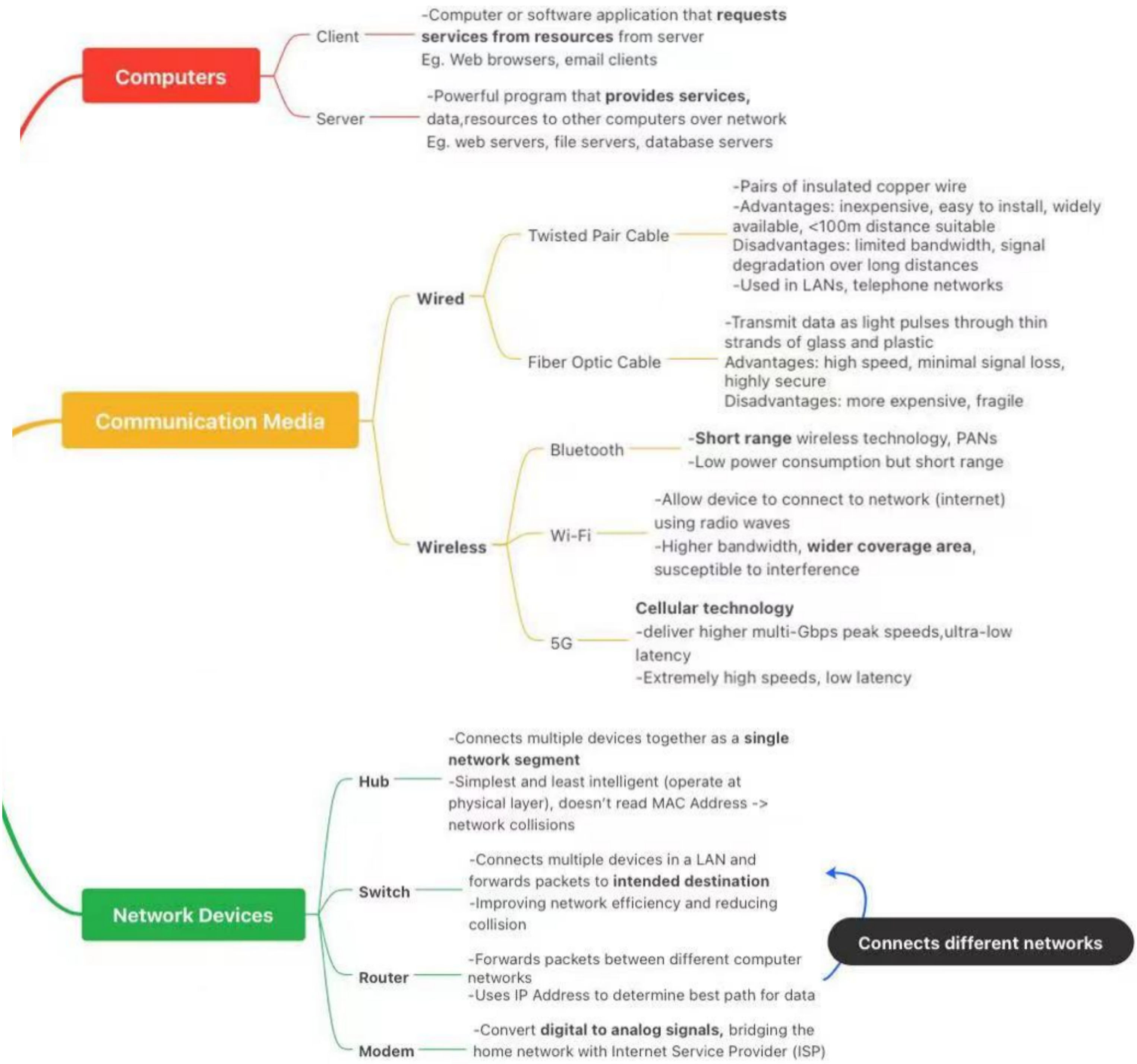
Range: Network, Database, Machine Learning

Note: (Review Guide/Review Practices/Practice Set) are UNOFFICIAL sources of reference and do NOT include or hint at questions in the actual exam!

Networks

- Interconnected computers and other devices that can share resources and information
- Advantages of networking (resource sharing, communication, centralized management)



IP (Internet Protocol) assigned to each device to connect to a computer network.
Functions:
-host or network interface
-location addressing

IP Address

IPv4: 32 bit numerical address
-four octets (0-255)
-classes (A,B,C,D,E)-> classful addressing allowed scaling of networks
-NID (Network ID) & HID (Host ID)

Class A (Large)
1st Octet: 0
2^24-2=16 million hosts (不能全0或1)
Default: 255.x.x.x
0.0.0.0-127.255.255.255

Class B (Medium)
1st Octet: 10
2^16-2=65,534 hosts
Default: 255.255.x.x
128.0.0.0-191.255.255.255

Class C (Small)
1st Octet: 110
2^8-2=254 hosts
Default: 255.255.255.x
192.0.0.0-223.255.255.255

Within network, host ID must be unique to network

Network Protocols

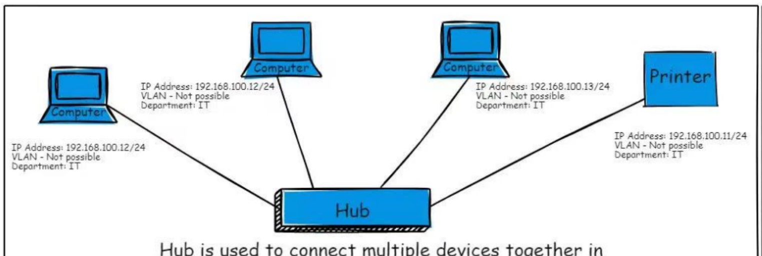
-Communication Protocols for addressing and routing data packets across networks

HTTP/ HTTPS (Hypertext Transfer Protocol)
-Transmitting hypertext over Internet

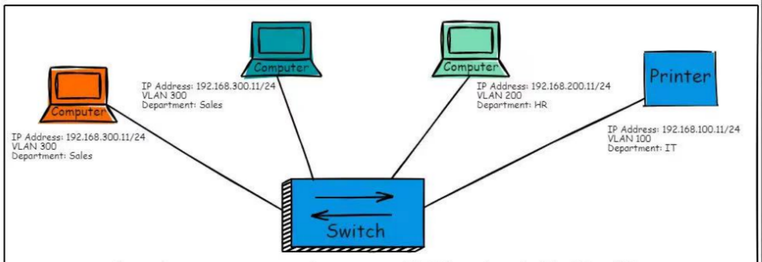
FTP (File Transfer Protocol)
-transferring computer files from server to client

SMTP (Simple Mail Transfer Protocol)
-Internet standard for mail transmission

Sets of rules that govern how data is formatted, transmitted, and received across network



Hub is used to connect multiple devices together in a single network (Broadcast Domain)



Switch can segregate devices via VLAN and each VLAN will have its own and separate Broadcast Domain

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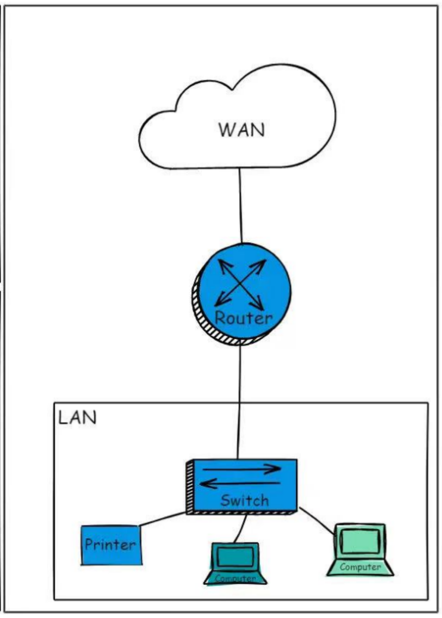


Database

- Organized, shared collection of data for long term use
- Database Management System (DBMS) is software used to manage databases for data definition, data manipulation, data control
- Database System (DBS) encompasses data, DBMS, application programs using the data and users interacting with it

Relational Database: most prevalent type of databases

- Store data in tables, and relationships between tables are established through common fields
- SQL (Structure Query Language)

Basic Definition Language

1. **INSERT INTO**: Inserts new rows

INSERT INTO Student (SID, Stu_Name, Gender, DOB, Nationality)

VALUES (1009, 'John', 'M', '2004-07-15', 'UK')

Ex:

INSERT INTO GFG_Employees (id, name, email, department)

VALUES
(1, 'Jessie', 'jessie23@gmail.com', 'Development'),

id	name	email	department
1	Jessie	jessie23@gmail.com	Development
2	Praveen	praveen_dagger@yahoo.com	HR
3	Bisa	dragonBall@gmail.com	Sales
4	Rithvik	msvv@hotmail.com	IT
5	Suraj	srjsunny@gmail.com	Quality Assurance
6	Om	OmShukla@yahoo.com	IT
7	Naruto	uzumaki@konoha.com	Development

GFG_Employees

(2, 'Praveen', 'praveen_dagger@yahoo.com', 'HR')

2. **UPDATE**: Updates existing data

UPDATE Student

SET column1= value1, column2=value2

WHERE Stu_Name = 'Bob' (condition)

3. **DELETE** FROM: Deletes data

DELETE FROM Student (table name)

WHERE Stu_Name = 'Frank' (some condition)

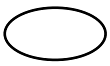
4. **SELECT**: Queries data

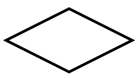
SELECT Stu_Name, Score

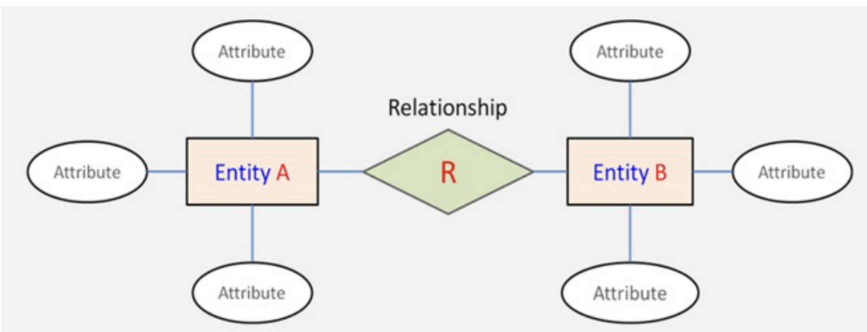
FROM Students

WHERE c.C_Name = 'Computer Science'

Entity-Relationship Diagram (ERD)

- Graphical tool used to represent entities, attributes, and relationships between entities
- Entities 
- Attributes 
- Relationships: relationship types (one-to one, one-to-many, many-to-many)





Machine Learning

- Enables computer systems to learn from data (discover patterns in large amounts of data -> make predictions)
- Supervised Learning: learning from labeled data, learns mapping relationship between input & outputs
- Tasks: Classification (predicting discrete class labels), Regression (predicting continuous numerical outputs)
- Unsupervised Learning: learn inherent structures, patterns, or associations within the data
- Tasks: Clustering (grouping data with high similarity), Dimensionality Reduction (PCA) (reduce number of data features while retaining relative positions accurate)

Common Algorithms

1. KNN

- Classification and regression
- Calculate distance between new sample and others in training set (Euclidean and Manhattan)
- Euclidean: $p = (p_1, p_2, \ldots, p_n)$ and $q = (q_1, q_2, \ldots, q_n)$ in n-dimensional

$$d(p,q) = \sqrt{\sum_{i=1}^{n}(p_i - q_i)^2}$$

  space
- Manhattan:
- Select K Nearest Neighbors: Identify the K training samples closest to the new sample

Detailed Explanation: https://www.youtube.com/watch?v=HVXime0nQeI

$$d(p,q) = \sum_{i=1}^{n} |p_i - q_i|$$

2. K-Means Algorithm

- Unsupervised learning clustering algorithm that aims to partition a dataset into K clusters
- Initialization: Randomly select K data points as initial cluster centroids
- Assign each data point to the cluster whose centroid is closest
- For each data point x_j, assign it to cluster C_i if $||x_j - \mu_i||^2 \leq ||x_j - \mu_k||^2 \quad \forall k \neq i$
- Recalculate the centroid of each cluster as the

mean of all data points assigned to that cluster

- Repeat steps 2 and 3 until the cluster centroids no longer change significantly

Detailed Explanation: https://www.youtube.com/watch?v=4b5d3muPQmA

3. PCA (Principal Component Analysis)

- Transform original data into a new set of orthogonal axes, called principal components, through a linear transformation

Detailed Explanation: https://www.youtube.com/watch?v=FgakZw6K1QQ&t=425s